

ConnectTel Customer Churn Prediction

MACHINE LEARNING INTERNSHIP PROJECT

MAY 2026

ConnectTel is losing millions in revenue every quarter as customers leave without warning. This project presents a machine learning system that predicts which customers will churn before they do — enabling the retention team to intervene with targeted, cost-effective offers. Built on the Telco Customer Churn Dataset (7,043 customers, 21 features), the best-performing model (XGBoost) achieves an AUC-ROC of 0.8470 and catches 79.7% of actual churners, with an estimated ROI of 3,471%.

Author: SN PRANAV | **Dataset:** Telco Customer Churn — Kaggle



Problem Statement

ConnectTel is losing millions in revenue every quarter due to customer churn. Acquiring a new customer costs **5 to 7 times more** than retaining an existing one — yet the retention team has no proactive way to identify at-risk customers before they leave.

26.5% Churn Rate

More than 1 in 4 customers leave every quarter — a critical and unsustainable loss.

\$120 Lost Per Customer

Each churned customer represents \$120 in lost monthly recurring revenue.

No Early Warning

The retention team cannot identify at-risk customers proactively — interventions happen too late.

 **Our Solution:** Build a machine learning system that predicts which customers will churn before they leave, enabling the retention team to intervene with targeted offers.

Dataset Overview

Dataset at a Glance

- **7,043** customers
- **21** features
- **26.5%** churn rate
- Moderately imbalanced

Source: Telco Customer Churn Dataset — Kaggle

Feature Categories

Demographics

Gender, Senior Citizen,
Partner, Dependents

Services

Phone, Internet, Streaming,
Security, Tech Support

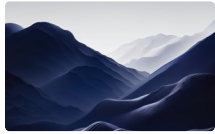
Account

Tenure, Contract Type, Payment Method, Monthly Charges

📌 **Key Data Issue Fixed:** `TotalCharges` was stored as a string with 11 blank entries for new customers. Converted to float and imputed with 0 — the correct business value representing no charges accrued yet.

EDA Key Findings

Exploratory Data Analysis revealed six statistically significant patterns that directly informed feature engineering and model design.



Fiber Optic vs DSL

Fiber Optic users churn at **41.9%** vs DSL at **18.9%** — a 23-point gap ($p < 0.05$).



Contract Type is #1 Driver

Month-to-month customers churn at **42.7%** vs only **2.8%** for two-year contracts — the single largest churn driver.



Tenure Effect

Customers in their first **12 months** have a 50%+ churn rate. After month 24, the rate drops significantly.



Partner Status

Customers without a partner churn at **32.9%** vs **19.7%** with a partner. Family ties reduce churn risk significantly.

Class Imbalance

73.5% loyal vs 26.5% churners. Accuracy is misleading — **AUC-ROC** used as the primary metric throughout.

Service Engagement

Customers with **4+ add-on services** churn at half the rate of single-service customers. Engagement is a strong retention signal.

Feature Engineering

Six new predictive features were engineered from raw columns to capture behavioral signals not directly available in the dataset.

1

TotalChargesPerTenure

$\text{TotalCharges} \div \text{tenure}$ — Captures average monthly spend; separates price-sensitive customers from high-value ones.

2

ServiceCount

Sum of 8 add-on service flags (0–8) — A direct proxy for customer engagement level.

3

HasFamilyTies

$\text{Partner} = \text{Yes OR Dependents} = \text{Yes}$ — Stability signal; family customers churn significantly less.

4

IsNewCustomer

$\text{Tenure} \leq 12 \text{ months}$ — Flags the highest-risk vulnerability window identified in EDA.

5

IsHighValue

MonthlyCharges above 75th percentile — High charges combined with month-to-month contract creates the highest-risk segment.

6

ContractRisk

Ordinal encoding: Month-to-month = 2, One year = 1, Two year = 0 — Preserves natural risk order for tree-based models.

Model Training Approach

Why Not Just Accuracy?

A model predicting "no churn" for everyone scores **73.5% accuracy** while being completely useless. **AUC-ROC** measures ranking quality independently of any threshold — the correct metric for imbalanced classification.

Handling Class Imbalance

- `class_weight=balanced` for Logistic Regression & Random Forest
- `scale_pos_weight=2.77` for XGBoost
- No SMOTE required — weighted approaches sufficient

Cross-Validation Strategy

StratifiedKFold with 5 splits preserves the 26.5% churn ratio in every fold. `StandardScaler` placed inside the Pipeline to prevent data leakage — fitted on training folds only.

Data Split

Stratified **80/20** train/test split. Both sets preserve the 26.5% churn ratio. `random_state=42` for full reproducibility.

Stratified Split

Build Pipeline

StratifiedKFold

Evaluate AUC-ROC

This end-to-end pipeline ensures no data leakage, preserves class distribution at every stage, and evaluates models on the most appropriate metric for imbalanced churn prediction.

Model Results

Three models were trained and evaluated on the held-out 20% test set. **XGBoost (tuned)** emerged as the winner with the highest AUC-ROC and the highest Recall — correctly identifying 79.7% of all actual churners.

Model	AUC-ROC	Recall	Precision	F1	Accuracy
Logistic Regression	0.8410	0.7914	0.4992	0.6122	0.7339
Random Forest	0.8442	0.7380	0.5691	0.6426	0.7821
XGBoost (tuned)	0.8470	0.7968	0.5129	0.6241	0.7452

GridSearchCV Best Parameters

Found across **120 total model fits** (24 combinations × 5 folds):

- `n_estimators: 300`
- `max_depth: 5`
- `learning_rate: 0.05`
- `subsample: 0.8`

Optimal Business Threshold

The default threshold of **0.50** is not optimal for churn. A business threshold of **0.37** maximizes revenue saved — balancing \$120 customer LTV against a \$25 retention offer cost per customer.

- ✓ At threshold 0.37: **633 customers** flagged vs 298 at default — catching 392 additional true churners.

SHAP Interpretability

SHAP converts XGBoost's internal calculations into plain feature contributions. A **positive SHAP value** pushes the prediction toward churn; a **negative value** pushes toward staying.

Top 5 Churn Drivers

1 ContractRisk — SHAP 0.692

Month-to-month contracts have SHAP values 2–3× higher than any other feature. The single largest driver.

2 Tenure — SHAP 0.461

Strong protective effect. SHAP values highly positive in the first 12 months, dropping sharply after month 24.

3 Internet Service: Fiber Optic — SHAP 0.310

Fiber Optic users consistently push predictions toward churn across the entire dataset.

4 Monthly Charges — SHAP 0.249

Near zero effect below \$55/month. Turns sharply positive above \$65. Effect amplified for Fiber Optic users.

5 Service Count — SHAP 0.141

Consistent negative values. Each additional add-on service measurably reduces churn risk.

Waterfall Plot Case Studies

High-Risk Churner

Month-to-month + 4-month
tenure + \$89/month → **87% churn
probability**

Safe Stayer

2-year contract + 58-month
tenure + 6 services → **6% churn
probability**

Borderline Case

Contract pushing toward churn,
tenure and services pushing back
→ **51% — model uncertain**

Business Recommendations

Five targeted interventions are recommended, each tied to a specific at-risk segment identified by the model. Combined, they protect an estimated **\$47,040/month** in revenue.

1. Lock-In Month-to-Month Customers

Target: Tenure <12 months, month-to-month contract

Action: Offer 15% discount to upgrade to 1-year contract

Impact: ~\$5,030/month protected

2. Fiber Optic Bundle Offer

Target: Fiber Optic users paying above \$65/month

Action: Free Online Security and Tech Support for 12 months

Impact: ~\$3,930/month protected

3. Onboarding Excellence Program

Target: All customers with tenure ≤12 months

Action: Structured check-ins at months 1, 3, 6, 9 + loyalty reward

Impact: ~\$9,432/month protected

4. Service Upsell Campaign

Target: ServiceCount ≤2 with month-to-month contract

Action: First 3 months free on 2 additional add-on services

Impact: Reduces churn rate by ~15%

5. Deploy Threshold 0.37 in Production

Target: All customers with predicted churn probability above 0.37

Action: Flag 633 customers for retention outreach (vs 298 at default)

Impact: Catches 392 true churners — estimated **ROI of 3,471%**

Cost-Benefit Summary & Next Steps

633

Customers Flagged

At threshold 0.37 vs 298 at default

392

True Churners Caught

Estimated actual churners identified

\$47K

Monthly Revenue Protected

Estimated across all campaigns

3,471%

Estimated ROI

Based on \$25 offer vs \$120 LTV

Limitations

- **Correlation vs causation** — SHAP shows which features the model uses, not why customers churn
- **Offer acceptance** — Revenue estimates assume 20–25% acceptance; A/B testing required
- **Model drift** — Customer behavior changes over time; retrain quarterly on fresh data

Next Steps

- Build a **Streamlit dashboard** for instant churn probability with SHAP explanation
- Deploy as a **REST API** (Flask/FastAPI) integrated with ConnectTel's CRM
- Design an **A/B test** to validate whether retention offers actually reduce churn

Tech Stack

Python · Pandas · NumPy · Scikit-learn · XGBoost · SHAP · Google Colab · GitHub

📄 **SN PRANAV** — Machine Learning Internship, May 2026 | Best Model: XGBoost (AUC-ROC 0.8470) | [GitHub Repository](#) | [Live Notebook](#)